Predicting Diabetes Using Machine Learning Techniques: A Comparative Study of KNN, SVM, Neural Networks, and Random Forests

Author:
Muhammad Reza Jafari

Course:
Data Science in Python

Submission Date:
24/06/2026

Contents

Tables:

Figures:

1. Introduction & Research Question

Diabetes is one of the most common chronic diseases worldwide and can lead to serious health complications if left undiagnosed. Early identification of individuals at risk can support preventive interventions and improve patient outcomes.

Machine learning techniques have increasingly been applied in healthcare to assist in disease prediction based on patient characteristics and medical measurements. This project investigates whether machine learning models can accurately predict diabetes using clinical indicators collected from patients.

Research Question

Which machine learning algorithm among KNN, SVM, Neural Network, and Random Forest provides the most reliable prediction of diabetes?

2. Dataset Description

This study uses the Pima Indians Diabetes Dataset, a well-known healthcare dataset frequently used for binary classification tasks.

Dataset URL: https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database

The dataset contains information collected from female patients of Pima Indian heritage and includes medical measurements that may be associated with diabetes.

Dataset Characteristics

- Number of observations: 768
- Number of features: 8
- Target variable: Outcome
    - 0 = No Diabetes
    - 1 = Diabetes

Table 1 Dataset Feature Description

Feature	Description
Pregnancies	Number of pregnancies
Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure
SkinThickness	Triceps skin fold thickness
Insulin	Serum insulin level
BMI	Body Mass Index
DiabetesPedigreeFunction	Family history indicator
Age	Age of patient

Outcome	Diabetes diagnosis

The dataset contains numerical variables only, making it suitable for machine learning classification methods.

3. Exploratory Data Analysis and Preprocessing

Initial Inspection

The dataset contains 768 observations and 9 columns. No missing values were formally recorded in the dataset.

However, several medical variables contained zero values that are physiologically unrealistic and likely represent missing measurements.

The dataset was first divided into training (80%) and testing (20%) subsets using stratified sampling. Unrealistic zero values in selected medical variables were subsequently treated as missing values. Missing values were imputed using the median calculated from the training data and then applied to both training and testing sets. This procedure prevented information leakage from the test set into the model training process.

Table 2 Number of Zero Values in Features' dataset

Variable	Zero Count
Glucose	5
BloodPressure	35
SkinThickness	227
Insulin	374
BMI	11

These values were treated as missing observations.

Missing Value Treatment:

The zero values were replaced with missing values (NaN). Missing values were subsequently imputed using the median of each feature because several variables exhibited skewed distributions.

Target Distribution:

The target variable was moderately imbalanced:

Table 3 Target Distribution

Outcome	Percentage
No Diabetes (0)	65.1%
Diabetes (1)	34.9%

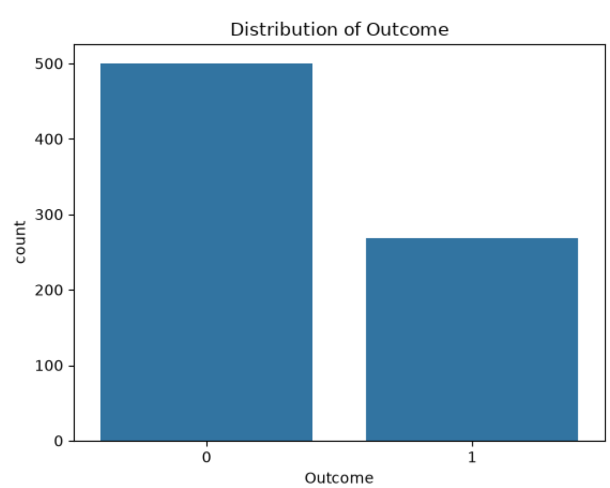

Figure 1 Distribution of outcome

Although some imbalance exists, the class distribution remains acceptable for supervised learning.

Feature Scaling:

Following the imputation of missing values, all predictor variables were standardized using StandardScaler, resulting in features with a mean close to 0 and a standard deviation close to 1.

Feature scaling was necessary because KNN, SVM, and Neural Network models are sensitive to differences in feature scales. Standardization ensured that all variables contributed equally during model training and improved the overall performance of the models.

Feature Relationships:

Correlation analysis showed that Glucose exhibited the strongest positive relationship with diabetes status. BMI and Age also demonstrated moderate associations with the target variable.

Histograms and boxplots revealed several skewed variables, particularly Insulin and SkinThickness, which justified the use of median imputation.

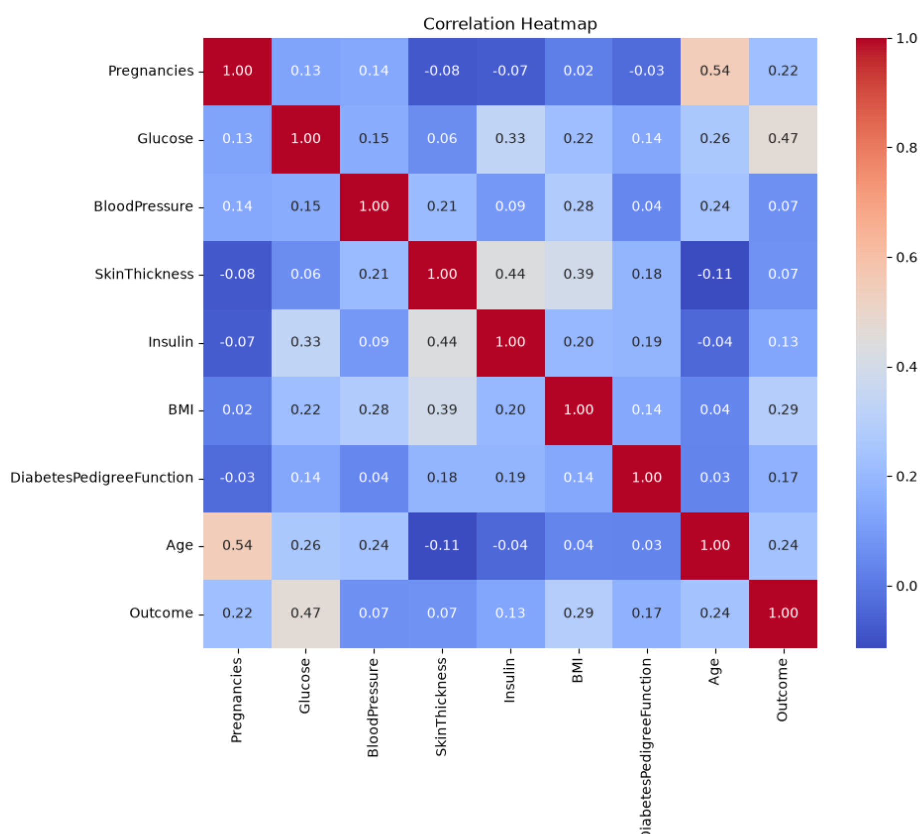

Figure 2 Correlation Heatmap

4. Methodology

Four machine learning classification algorithms were evaluated:

Random Forest

Random Forest is an ensemble learning method that combines multiple decision trees built on bootstrap samples of the training data. By aggregating predictions from many trees, Random Forest typically reduces overfitting and improves predictive performance compared with a single decision tree.

Hyperparameters tuned:

- Number of trees (n_estimators)
- Maximum tree depth (max_depth)

K-Nearest Neighbors (KNN)

KNN classifies observations based on the majority class among the nearest neighboring observations. Hyperparameters were optimized using GridSearchCV.

Tuned parameters:

- Number of neighbors (k)
- Weighting scheme

Support Vector Machine (SVM)

SVM attempts to identify an optimal decision boundary separating diabetic and non-diabetic patients.

Tuned parameters:

- Kernel type
- Regularization parameter (C)
- Gamma

Neural Network (MLP)

A Multi-Layer Perceptron (MLP) classifier was implemented to model nonlinear relationships within the data.

Tuned parameters:

- Hidden layer structure
- Regularization parameter (alpha)

Model Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Ten-fold cross-validation was used during hyperparameter optimization.

5. Results

Table 4 Evaluation Table

Model	Accuracy	Precision	Recall	F1	ROC-AUC
KNN	0.7532	0.6818	0.5556	0.6122	0.8100
SVM	0.7273	0.5938	0.7037	0.6441	0.8065
Neural Network	0.7727	0.6863	0.6481	0.6667	0.8337
Random Forest	0.7338	0.6585	0.5000	0.5684	0.8139

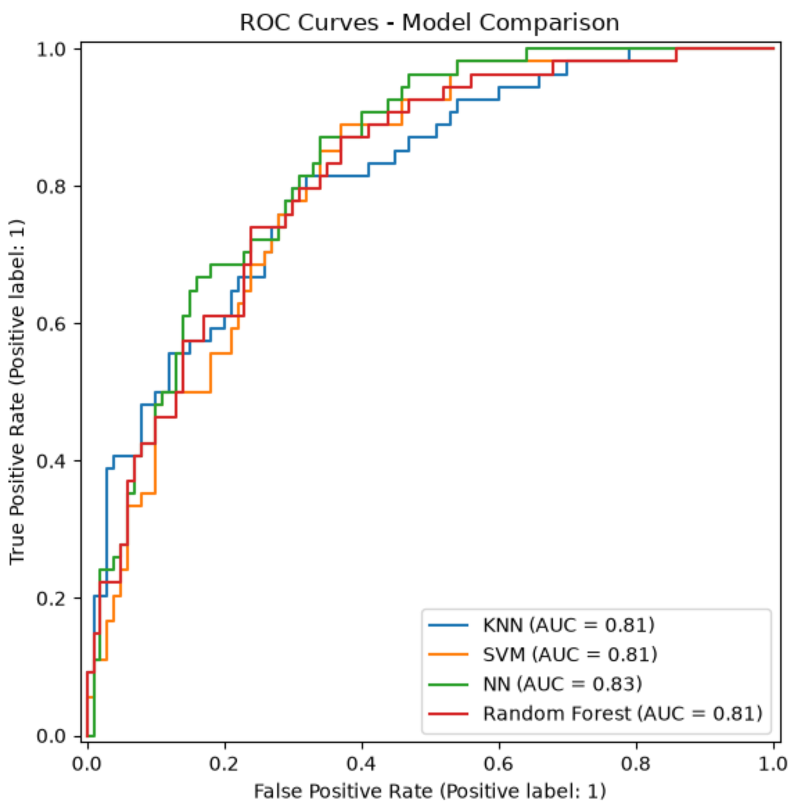

Figure 3 ROC Curves- Model Comparison

7. Results and Discussion

Four machine learning algorithms were evaluated for diabetes prediction: K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Neural Network (MLP), and Random Forest.

The performance of each model is summarized in Result Table section.

The Neural Network achieved the best overall performance, obtaining the highest Accuracy (77.3%), Precision (68.6%), F1-score (66.7%), and ROC-AUC (83.4%). These results indicate that the Neural Network provided the most balanced and reliable predictions across multiple evaluation metrics, if you have more data the performance of Neural Network would be even better because it better with huge amount of data.

The SVM model achieved the highest Recall (70.4%), meaning that it identified the largest proportion of diabetic patients. From a healthcare perspective, Recall is particularly important because missed diabetes cases may delay diagnosis and treatment. However, the improved Recall of SVM came at the cost of lower Precision and Accuracy.

KNN demonstrated competitive performance but was outperformed by the Neural Network on all major metrics except Precision, where the difference was negligible. Random Forest produced acceptable results but showed the lowest Recall, indicating that it missed a larger number of diabetic patients than the other models.

Overall, the Neural Network appears to be the most suitable model for this dataset because it achieved the strongest balance between discrimination ability, predictive accuracy, and robustness.

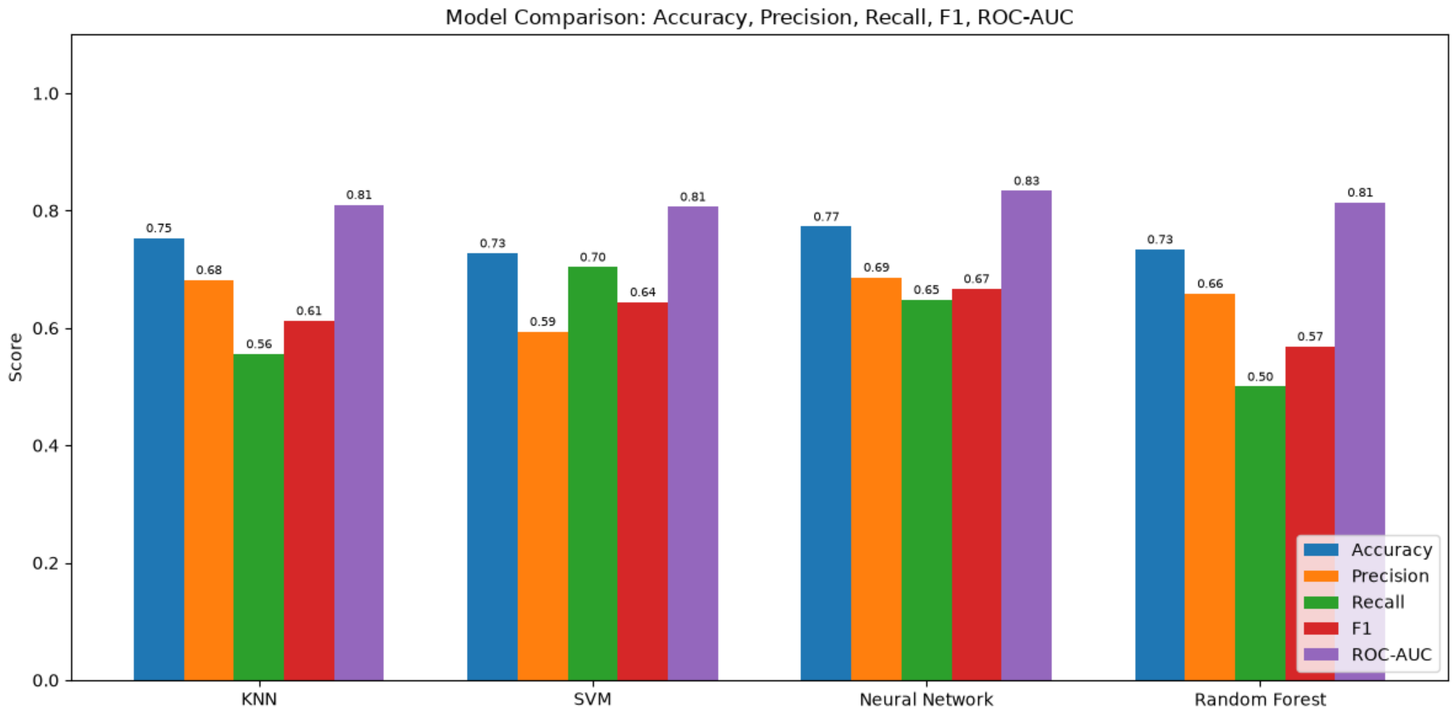

Figure 4 All Model Comparison

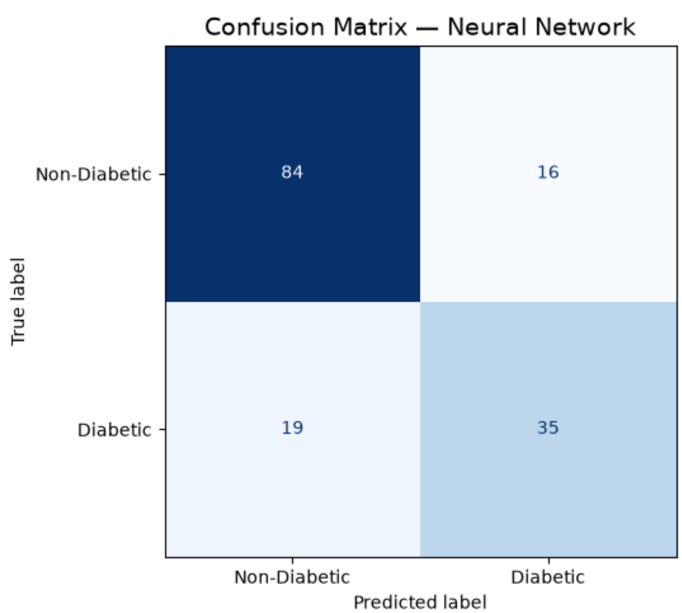

Figure 5 Confusion Matrix - Neural Network

8. Limitations

Several limitations should be acknowledged.

First, the dataset contains only 768 observations, which limits the complexity of patterns that can be learned. Second, the data originate from a specific population group and may not generalize to broader populations. Third, several medical measurements contained unrealistic zero values that required imputation, introducing some uncertainty into the analysis.

Future work could explore larger and more diverse healthcare datasets, advanced ensemble techniques such as Gradient Boosting, and feature engineering approaches to further improve predictive performance.

9. Conclusion

This project investigated the use of machine learning methods for diabetes prediction using the Pima Indians Diabetes Dataset. After performing exploratory data analysis, data cleaning, imputation, standardization, and hyperparameter optimization, four classification models were evaluated: KNN, SVM, Neural Network, and Random Forest.

Among the evaluated models, the Neural Network achieved the strongest overall performance, with an Accuracy of 77.3% and a ROC-AUC score of 83.4%. While SVM achieved the highest Recall, the Neural Network provided the best balance between correctly identifying diabetic patients and minimizing false predictions.

The findings demonstrate that machine learning techniques can effectively support diabetes risk prediction and may serve as useful tools for assisting healthcare professionals in early screening and decision-making.

References

- UCI Machine Learning Repository. Pima Indians Diabetes Dataset.
- Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.
- Kaggle. Pima Indians Diabetes Database. https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database